Analytics Pipeline Architecture

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1. Overview

The Analytics Pipeline (analytics-pipeline) is owned by the Data Platform team, with Emma Larsen as technical lead. It is implemented in Python and PySpark and listens on port 8700. Database schema migrations must be backward compatible for at least one release. We should revisit the alert thresholds after the next game day. Remember to tag pull requests with the owning team so reviews get routed correctly.

Feature work is paused during incident follow-up weeks until action items are closed. All services must expose Prometheus metrics on the /metrics path. Secrets must never be committed; the pre-commit hook scans for high-entropy strings. Dependency upgrades are batched weekly by the platform bot.

2. Data Stores

Primary data store: Snowflake warehouse ACME_ANALYTICS. Dependency upgrades are batched weekly by the platform bot. Structured JSON logging is mandatory; free-text logs are dropped by the log pipeline.

3. Dependencies

Analytics Pipeline depends on: Kafka, Airflow, Snowflake. Every service has a README that links to its dashboards and runbook. The dashboards were updated to show saturation alongside latency and errors.

4. Security and Authentication

Authentication mechanism: Analytics Pipeline uses service accounts with Snowflake key-pair authentication. All services must expose Prometheus metrics on the /metrics path. Every service has a README that links to its dashboards and runbook.

5. Scaling

We should revisit the alert thresholds after the next game day. Every service has a README that links to its dashboards and runbook. Service-level objectives are reviewed quarterly with product owners. Load tests should be coordinated in the platform channel so they do not trip alerts. Infrastructure changes go through Terraform plans that need two approvals.

Secrets must never be committed; the pre-commit hook scans for high-entropy strings. Database schema migrations must be backward compatible for at least one release. Dependency upgrades are batched weekly by the platform bot. Capacity planning for the holiday peak starts in September every year.

6. Open Questions

Structured JSON logging is mandatory; free-text logs are dropped by the log pipeline. Infrastructure changes go through Terraform plans that need two approvals. Structured JSON logging is mandatory; free-text logs are dropped by the log pipeline.